

Sec 4.10 ~ 12 perfectly universal sets of quantum gates

	classical	Quantum
state	$\{0,1\}^n$	$\mathbb{H}_n, \mathbb{C}^{2^n}$

Need to implement	Boolean ft.	Unitary matrix
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Universal set	$\{\text{AND}, \text{OR}, \text{NOT}\}$	?
	$\{\text{NAND}\}$	
	$\{\text{NOR}\}$	

$\{\text{H}, \text{T}, \text{CNOT}\}$

$\{\text{all rotation}, \text{CNOT}\}$

"primitive set"

Review

One qubit unitary operator $U \in \mathbb{H}_1$

$$U = e^{i\delta} A \otimes B \otimes C, \quad ABC = I$$

[Thm 4.3.3]

Def 4.8.4

(1) S : universal for a set T of unitary operators

if $\forall \epsilon \in \mathbb{R}_+$, $\forall U \in T$, $\exists$ unitary operator V .

$$E(U, V) < \epsilon.$$

$$E(U, V) = \sup_{\substack{|\varphi\rangle \in \mathbb{H}_n \\ \langle \varphi | \varphi \rangle = 1}} \|U|\varphi\rangle - V|\varphi\rangle\|$$

(2) S : universal for quantum computation

if S : universal for all unitary operators.

(3) Perfectly universal for quantum computation

If all unitary operators $U \in \mathcal{U}_n$ can be

implemented by the gates in S , auxiliary, erasure.

§ 4.10 Perfectly universal sets of quantum gates

Def 4.10.1, $A \in \mathbb{C}^{k \times k}$: two level gate.

if there are $i, j \in \mathbb{Z}_k$, s.t for $\forall \hat{v} \in \mathbb{C}^k$,

$$\hat{v}_k = A \hat{v}_k \quad \text{for } k \neq i, j$$

[2개 다른 보일
다 같으면]

Ex :

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & a & 0 & b \\ 0 & 0 & 1 & 0 \\ 0 & c & 0 & d \end{bmatrix} : \text{two-level gate}$$

$$U \in \mathbb{C}^{1 \times 1} ; \text{two-level gate.}$$

$$U \in \mathbb{C}^{2 \times 2}$$

Thm 4.10.4, Let $U \in \mathbb{C}^{k \times k}$ unitary,

Then U can be written as a product of $\frac{k(k-1)}{2}$

unitary two-level gates, for $k \geq 2$.

Proof),

Induction

$n=2$ $U \in \mathbb{C}^{2 \times 2}$: two-level gate.

Assume. $n=k$ hold. : For $U \in \mathbb{C}^{k \times k}$

$$U_k = \underbrace{V_{\frac{k(k-1)}{2}} \cdots V_1}_{\frac{k(k-1)}{2} \text{ product.}} \quad \text{where } V_j: \text{ two level gate}$$

For $n=k+1$ consider $U_{k+1} \in \mathbb{C}^{(k+1) \times (k+1)}$,

sketchy, [we want to find $\frac{(k+1)k}{2}$ two level gate]

$$U_{k+1} = A_k \cdots A_1 \begin{pmatrix} 1 & 0 \\ 0 & U_k \end{pmatrix} = A_k \cdots A_1 \bar{V}_{\frac{k(k-1)}{2}} \cdots \bar{V}_1$$

where $\bar{V}_k := \begin{pmatrix} 1 & 0 \\ 0 & V_k \end{pmatrix} \in \mathbb{C}^{(k+1) \times (k+1)}$

Claim (two-level gate can be ~~approx~~ $\begin{pmatrix} 1 & 0 \\ 0 & V_k \end{pmatrix}$ ~~is~~),

$$U_{k+1} = A_k \cdots A_1 \begin{pmatrix} 1 & 0 \\ 0 & V_k \end{pmatrix},$$

two level gate in $\mathbb{C}^{(k+1) \times (k+1)}$

Construction of A_k .

$$U_{k+1} = (a_{ij})_{(k+1) \times (k+1)}$$

$$C = \frac{1}{\sqrt{|a_{11}|^2 + |a_{k+1,1}|^2}}$$

$$A_k = \begin{bmatrix} C \bar{a}_{11} & 0 & \cdots & 0 & C \bar{a}_{k+1,1} & 0 & \cdots & 0 \\ 0 & & & & 0 & & & \\ \vdots & & I & & \vdots & & & \\ 0 & & & & 0 & & & \\ k\text{th row} & C a_{k,1} & 0 & \cdots & 0 & -C a_{1,1} & & \\ 0 & & & & & & & \\ \vdots & & & & & & I & \\ 0 & & & & & & & \end{bmatrix}$$

: Unitary two column gates.

$$(A_k U_{k+1})_{(1,1)} = C(a_{11})^2 + C(a_{k,1})^2 = \frac{1}{C}$$

$$(A_k U_{k+1})_{(1,k)} = C a_{k1} a_{11} - C a_{11} a_{k1} = 0.$$

$$\underbrace{A_1 \dots A_k U_{k+1}}_{\text{Product of unitary}} = \begin{bmatrix} x_1 & * & \dots & * \\ 0 & * & \dots & * \\ \vdots & \vdots & & \vdots \\ 0 & * & \dots & * \end{bmatrix}.$$

$\Rightarrow$ Unitary,

column : orthonormal $\Rightarrow$

basis $x=1$

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & * & \\ 0 & & & \end{bmatrix},$$

We proved

$$U_{k+1} = A_k^{-1} \dots A_1^{-1} \begin{bmatrix} 1 & 0 \\ 0 & U_k \end{bmatrix}.$$

Cor: Set of all two level gates is

perfectly universal for quantum computation

Thm 4.10.6, $S := \{ \text{all rotation gates,} \\ \text{standard CNOT gate} \}$

perfectly universal for quantum computing

Def 4.10.7, Let $\vec{s}, \vec{t} \in \{0,1\}^n$.

A Gray code connecting $\vec{s}$ and $\vec{t}$ is a sequence.

$G = (\vec{g}_1, \dots, \vec{g}_m)$ pairwise distinct vectors in $\{0,1\}^n$.
" " "
 $\vec{s}$ $\vec{t}$ and successive elements G differ
by exactly one bit.

Example. Gray code $\vec{s} = (0,0,0)$ $\vec{t} = (1,1,1)$

$G = ((0,0,0), (1,0,0), (1,1,0), (1,1,1))$

Prop. 4.10.10. Let $\vec{s}, \vec{t} \in \{0,1\}^n$, then there is

a Gray code of length $\leq n+1$ that connects $\vec{s}$ and $\vec{t}$.

Thm 4.10.12.

For every two-level unitary gate U on $\mathbb{H}_n$,

$\exists$ unitary single qubit operator V .

s.t. U can be implemented by quantum circuits

that uses V , $O(n^2)$ Pauli X , Hadamard, $Z/8$,

standard CNOT, ancillary, erasure gates,

and four other single qubit gates.

Proof)

U : two level gate. on $\mathbb{H}_n$,

For $\vec{s}, \vec{t} \in \{0, 1\}^n$, $\alpha, \beta, r, \delta \in \mathbb{C}$

$$U|\vec{s}\rangle = \alpha|\vec{s}\rangle + \beta|\vec{t}\rangle, \quad U|\vec{t}\rangle = r|\vec{s}\rangle + \delta|\vec{t}\rangle,$$

U leaves all other computational basis. Ex $\begin{bmatrix} 1 & & \\ & \alpha & \beta \\ & r & \delta \end{bmatrix}$

define $V \in \mathbb{H}_1$.

$$V|0\rangle = \alpha|0\rangle + \beta|1\rangle, \quad V|1\rangle = r|0\rangle + \delta|1\rangle$$

Claim $\exists p$: unitary s.t

$$p|\vec{s}\rangle = |t_0 \dots t_{i-1}\rangle |0\rangle |t_{i+1} \dots t_{n-1}\rangle$$

$$p|\vec{t}\rangle = |t_0 \dots t_{i-1}\rangle |1\rangle |t_{i+1} \dots t_{n-1}\rangle$$

If the claim hold.

$$U = P^\dagger C^{G, G_1, i}(V) P,$$

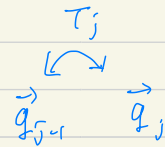
U is implemented by $P^\dagger, C^{G, G_1, i}(V)$

Proof of the claim

Let G : gray code. $G = (\vec{g}_1, \dots, \vec{g}_m)$

Let T_j : transposition operator

$$s, t, \quad |\vec{g}_t\rangle = T_j(|\vec{g}_s\rangle), \quad |\vec{g}_{s-1}\rangle = T_j(|\vec{g}_s\rangle)$$



$$\text{Let } P^\dagger = T_{m-1} \dots T_1$$

$$\Rightarrow P(|\vec{s}\rangle) = |\vec{g}_{sm+1}\rangle, \quad T_m(|\vec{g}_{m-1}\rangle) = |\vec{g}_m\rangle = |\vec{t}\rangle,$$

$P(|\vec{s}\rangle)$ and $|\vec{t}\rangle$ differ in exactly one qubit

If $t_i = 1$, claim is prove.

If $t_i = 0$ replace P by $\text{Trans}_{\vec{t}} P$, ($\vec{t} = (t_0 \dots t_{i-1} \neq t_{i+1} \dots t_{m-1})$).

P : $O(n)$ transposition: $X, H, T, CNOT$, ancilla, erasure.
(Thm 4.9.15)

$C^{G, G_1, i}(V)$: $X, H, T, CNOT$, ancilla, erasure

(Thm 4.9.14)

Goal : Finding Universal set

for the quantum computing

$\{ U \in \mathbb{H}_n \}$: Universal

Thm 4.10.4.

$\{ U \in \mathbb{H}_n : \text{two-level gate} \}$: Universal.

Thm 4.10.6

$\left\{ \begin{array}{l} U \in \mathbb{H}_1 : \text{unitary single qubit operator} \\ \text{X, T, H, CNOT} \end{array} \right\}$: Universal

rotation : unitary single qubit

$\{ U \in \mathbb{H}_1, \text{CNOT} \}$: universal

$\{ H, T, \text{CNOT} \}$: universal,

" " (T/8 gate)

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix}$$

$\left(\{ H, T \} : \text{universal for } U \in \mathbb{H}_1 \right)$ Thm 4.11.2.

Sketch of the proof of Thm 4.11.2

$\{H, T\}$: universal for $V \in \mathcal{H}_1$,

$$\text{Let } \hat{n} = \frac{1}{\sqrt{\cos^2 \frac{\pi}{8} + 1}} \left(\cos \frac{\pi}{8}, \sin \frac{\pi}{8}, \cos \frac{\pi}{8} \right)$$

$$\hat{m} = \frac{1}{\sqrt{\cos^2 \frac{\pi}{8} + 1}} \left(\cos \frac{\pi}{8}, -\sin \frac{\pi}{8}, \cos \frac{\pi}{8} \right)$$

$\hat{n}, \hat{m}$: non parallel unit vector

By Thm 4.3.35. $\exists \delta, d_i, \beta_i$ for $0 \leq i \leq k-1$ s.t.

$$V = e^{i\delta} \prod_{i=0}^{k-1} R_{\hat{n}}(d_i) \underbrace{R_{\hat{m}}(\beta_i)}$$

by Lem 4.11.4

$$= e^{i\delta} \prod_{i=0}^{k-1} \underbrace{R_{\hat{n}}(d_i)} \quad \underbrace{R_{\hat{m}}(\beta_i)} \quad \text{H.}$$

For each rotation

Proposition 4.11.10- (2)

For all $\epsilon > 0$. $\exists a_i, b_i \in \mathbb{N}$ s.t.

$$E(R_{\hat{n}}(d_i), R_{\hat{n}}^{a_i}(0)) < \frac{\epsilon}{2k}.$$

$$E(R_{\hat{m}}(\beta_i), R_{\hat{m}}^{b_i}(0)) < \frac{\epsilon}{2k}.$$

$$\text{Let } V = \prod_{i=0}^{k-1} R_{\hat{n}}^{a_i}(\theta) H R_{\hat{n}}^{b_i}(\theta) H.$$

Prop 4.11.3 (3)

$$E(\prod_{k=1}^n U_k, \prod_{k=1}^n V_k) \leq \sum E(U_k, V_k)$$

$$E(U, V) \leq \sum_{i=0}^{k-1} E(R_{\hat{n}}^{a_i}(t_i), R_{\hat{n}}^{a_i}(\theta)) + \sum_{i=0}^{k-1} E(R_{\hat{n}}^{b_i}(\beta_i), R_{\hat{n}}^{b_i}(\theta))$$

$$\leq k \frac{\epsilon}{2k} + k \cdot \frac{\epsilon}{2k} \leq \epsilon.$$

$\Rightarrow$ we found approximation V .

$$\text{Lem 4.11.5 } R_{\hat{n}}(\theta) = e^{-i \frac{\pi}{4}} T H T H, \quad (4)$$

$$V = e^{-i \frac{\pi}{4} (\sum (a_i + b_i))} \prod_{i=0}^{k-1} (T H T H)^{a_i} H (T H T H)^{b_i} H.$$

Using $\{T, H\}$ we constructed V . s.t. $E(U, V) < \epsilon$

Proof of Prop 4.1.3) $E\left(\prod_{i=1}^k U_k, \prod_{i=1}^k V_k\right) \leq \sum_{i=1}^k E(U_k, V_k)$

Induction

$k=1$ Trivial

Let $k-1$ step hold.

i.e. $U = \prod_{i=1}^{k-1} U_i$ $V = \prod_{i=1}^{k-1} V_i$ $(k-1)$ product.

$$E(U, V) \leq \sum_{i=1}^{k-1} E(U_i, V_i)$$

For k -step, $U_k V$, $V_k V$: k product

$$E(U_k V, V_k V) = \sup_{\substack{\psi \in \mathcal{H}_n \\ \|U_k \psi\|=1}} \|(U_k V - V_k V) \psi\|_{\ell^1}.$$

$$\leq \sup_{\psi} \|U_k (V - V) \psi\|_{\ell^1} + \sup_{\psi} \|(U_k - V_k) V \psi\|_{\ell^2}.$$

$$\leq \sup \| (V - V) \psi \|_{\ell^2} + \sup \| (U_k - V_k) V \psi \|_{\ell^1}.$$

$\because U_k$ unitary

$$\leq E(U, V) + E(U_k, V_k) \leq \sum_{i=1}^k E(U_i, V_i)$$

Proof of Lem 4.11.4.) $R_{\hat{n}}(r) = H R_{\hat{p}}(r) H$

Def 4.3.4

$$R_{\hat{n}}(r) = e^{-i r \hat{n} \cdot \sigma / 2} \quad (\sigma = (X, Y, Z))$$

$$= \cos \frac{r}{2} I - i \sin \frac{r}{2} \hat{n} \cdot \sigma$$

$$= \cos \frac{r}{2} I - i \sin \frac{r}{2} (n_x X + n_y Y + n_z Z).$$

Note. $HH = I$, $HXH = Z$, $HYH = -Y$, $HZH = X$.

$$\begin{aligned} H R_{\hat{n}}(r) H &= \cos \frac{r}{2} I - i \sin \frac{r}{2} (n_x Z - n_y Y + n_z X), \\ &= R_{\hat{m}}(r) \end{aligned}$$

$\nwarrow \begin{matrix} \text{"} \\ n_z \end{matrix} \quad \overline{\begin{matrix} \text{"} \\ m_y \end{matrix}} \quad \begin{matrix} \text{"} \\ m_x \end{matrix}$

$$\left(\begin{array}{l} \because n_x = n_z = m_x = m_z = \cos \frac{\pi}{8}, \\ n_y = \sin \frac{\pi}{8} = -m_y \end{array} \right)$$

Proof of Lem 4.11.5. statement

$$\left(R_{\hat{n}}(\theta) = e^{-i \frac{\theta}{2} \hat{n} \cdot \sigma} \right)$$

Recall Def 4.3.4 $\theta = \arccos(\cos^2 \frac{\pi}{8})$

$$R_{\hat{n}}(\theta) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \hat{n} \cdot \sigma$$

$\cos \frac{\theta}{2} = \cos^2 \frac{\pi}{8}$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix} = e^{i\frac{\pi}{8}} \begin{bmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{bmatrix}$$

$\sin^2 \frac{\pi}{8} = 1 - \cos^2 \frac{\pi}{8}$
 $= \frac{1 - \cos^2 \frac{\pi}{8}}{1 + \cos^2 \frac{\pi}{8}}$
 $= \frac{1}{\| \hat{n} \|^2} \sin^2 \frac{\theta}{2}$

$$e^{-i\frac{\pi}{8}} T = \begin{bmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{bmatrix} = \cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} Z$$

$$e^{-i\frac{\pi}{8}} HTH = e^{-i\frac{\pi}{8}} H(|0\rangle\langle 0| + |1\rangle\langle 1|) H$$

$$= \cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} X$$

$$e^{-i\frac{\pi}{4}} THTH = \underbrace{e^{-i\frac{\pi}{8}} T}_{Z \otimes \frac{\pi}{4}} \underbrace{e^{-i\frac{\pi}{8}} THT}_{X \otimes \frac{\pi}{4} \text{ rotates}}$$

$$= \left(\cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} Z \right) \left(\cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} X \right)$$

$$= \cos^2 \frac{\pi}{8} I - i \sin \frac{\pi}{8} \cos \frac{\pi}{8} (X + Z) - \sin^2 \frac{\pi}{8} ZX$$

$\hat{n} \cdot \sigma$

$$= \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \left(\frac{1}{\| \hat{n} \|} \left(\cos \frac{\pi}{8} X + \sin \frac{\pi}{8} Y + \cos \frac{\pi}{8} Z \right) \right)$$

$$= R_{\hat{n}}(\theta)$$

Proof of Proposition 4.11.10.

$$\left(\begin{array}{l} \text{For } \forall \epsilon > 0, \forall r \in \mathbb{R}. \exists k \in \mathbb{N} \text{ s.t.} \\ E(R_{\hat{n}}(r), R_{\hat{n}}^k(\theta)) < \epsilon. \end{array} \right)$$

choose $r \in [0, 2\pi)$.

Note $\underbrace{\theta_k := k\theta}_{\text{countably many}} \pmod{2\pi}$ for $k \in \mathbb{N}$.

By Lem 4.11.9, $\exists k \in \mathbb{N}$ s.t. $|r - \theta_k| < \epsilon$.

For $|\psi\rangle \in \mathbb{H}$,

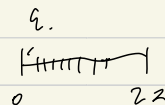
$$\| (R_{\hat{n}}(r) - R_{\hat{n}}^k(\theta)) |\psi\rangle \|_2.$$

$$= \| (\cos \frac{r}{2} - \cos \frac{\theta_k}{2}) I |\psi\rangle - i (\sin \frac{r}{2} - \sin \frac{\theta_k}{2}) \hat{n} \cdot \sigma |\psi\rangle \|_2.$$

$$\left(\text{Def } R_{\hat{n}}(\theta) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \hat{n} \cdot \sigma \right)$$

$$\leq 2 \left| \frac{r}{2} - \frac{\theta_k}{2} \right| < \epsilon.$$

Lem 4.11.9,



$\{k\theta \pmod{2\pi} : k \in \mathbb{N}\}$ is dense in $[0, 2\pi)$

$$\text{Let } I_k := \left[\epsilon_k, \epsilon(k+1) \right] \quad \left[\frac{2\pi}{\epsilon}, 1 \right] \quad \sum_{k=0}^{\left[\frac{2\pi}{\epsilon} \right]} I_k \supset [0, 2\pi)$$

$\Rightarrow \exists k, l$ s.t. $0 < k\theta - l\theta < \epsilon$ (pigeon hole principle)

$x \in [0, 2\pi) \Rightarrow \exists n, x = \epsilon n + (x - \epsilon n), \quad (x - \epsilon n) < \epsilon.$

$\exists v \in \mathbb{N}, \quad |v(k\theta - l\theta) - x| < \epsilon.$